

BM25 & TF–IDF Scoring, Worked Out by Hand

Keyword search ranks documents by BM25: rare terms count more, repeats saturate, long documents are penalised. We score one document for a two-word query, term by term.

What this document does. Before vectors, retrieval was lexical: match query *words* and weight them well. BM25 is the standard. This document builds a four-document corpus, derives the inverse-document-frequency weight for each query term, and computes the full BM25 score of one document — showing how rarity, term-frequency saturation, and length normalisation each move the number.

Concepts covered: term frequency · document frequency · inverse document frequency (idf) · the BM25 saturation term (k_1) · length normalisation (b) · summing term contributions.

Internal learning material. Numbers reproducible from the appendix code. v1.0

1 The setup: a four-document corpus

doc	contents (tokens)	length $ d $
d_1	database error in the database connection	6
d_2	payment error logged	3
d_3	user error on submit	4
d_4	cache warm complete	3

Corpus size $N = 4$. Average document length $\text{avgdl} = (6 + 3 + 4 + 3)/4 = 4.0$. The query is “database error”. We score d_1 .

Intuition

Two instincts drive lexical scoring. First, a *rare* query word (“database”, in one doc) is far more informative than a *common* one (“error”, in three) — match the rare word and you’ve found something specific. Second, the tenth occurrence of a word says little the first didn’t, and a long document hits any word more by sheer size. BM25 encodes all three: rarity reward, saturation, length penalty.

2 Step 1 — Document frequency and idf

The document frequency $\text{df}(t)$ is the number of documents containing term t :

$$\text{df}(\text{database}) = 1 \text{ } (d_1 \text{ only}), \quad \text{df}(\text{error}) = 3 \text{ } (d_1, d_2, d_3).$$

BM25’s inverse-document-frequency weight is

$$\text{idf}(t) = \ln\left(1 + \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5}\right).$$

$$\text{idf}(\text{database}) = \ln\left(1 + \frac{4 - 1 + 0.5}{1 + 0.5}\right) = \ln\left(1 + \frac{3.5}{1.5}\right) = \ln(3.333) = 1.204,$$

$$\text{idf}(\text{error}) = \ln\left(1 + \frac{4 - 3 + 0.5}{3 + 0.5}\right) = \ln\left(1 + \frac{1.5}{3.5}\right) = \ln(1.429) = 0.357.$$

The rare term “database” is weighted 1.204 — about $3.4\times$ the common “error” at 0.357.

Mini-example — this operation in isolation

idf falls as a term spreads across the corpus. With $N = 4$: a term in 1 doc scores $\ln(1 + 3.5/1.5) = 1.204$; in 2 docs, $\ln(1 + 2.5/2.5) = \ln 2 = 0.693$; in all 4, $\ln(1 + 0.5/4.5) = \ln(1.111) = 0.105$. A word everyone uses carries almost no signal — which is the whole point of *inverse* document frequency.

3 Step 2 — The BM25 term score

For each query term t with frequency $f = \text{tf}(t, d)$ in document d , BM25 adds

$$\text{score}(t, d) = \text{idf}(t) \cdot \frac{f(k_1 + 1)}{f + k_1(1 - b + b \frac{|d|}{\text{avgdl}})},$$

with standard constants $k_1 = 1.5$ (saturation) and $b = 0.75$ (length normalisation). For d_1 : $|d| = 6$, $\text{avgdl} = 4$, so $|d|/\text{avgdl} = 1.5$ and the length factor is

$$1 - b + b \frac{|d|}{\text{avgdl}} = 1 - 0.75 + 0.75 \cdot 1.5 = 0.25 + 1.125 = 1.375, \quad k_1 \cdot 1.375 = 2.0625.$$

Term “database” ($f = 2$ in d_1):

$$\frac{2(1.5 + 1)}{2 + 2.0625} = \frac{5.0}{4.0625} = 1.231, \quad \text{score} = 1.204 \times 1.231 = 1.482.$$

Term “error” ($f = 1$ in d_1):

$$\frac{1(2.5)}{1 + 2.0625} = \frac{2.5}{3.0625} = 0.816, \quad \text{score} = 0.357 \times 0.816 = 0.291.$$

4 Step 3 — Sum the terms

BM25 of a document is the sum over query terms:

$$\text{BM25}(d_1) = 1.482 + 0.291 = \boxed{1.773}.$$

term	df	idf	f in d_1	contribution
database	1	1.204	2	1.482
error	3	0.357	1	0.291
BM25(d_1)				1.773

Even though “error” appears, “database” provides 84% of the score — the rare term dominates, exactly as idf intends.

5 Step 4 — The two knobs, felt

Saturation (k_1). The factor $\frac{f(k_1+1)}{f+...}$ rises with f but flattens — doubling a word’s count never doubles its score:

f	1	2	5	10
$\frac{f \cdot 2.5}{f+2.0625}$	0.816	1.231	1.770	2.073

From $f = 1$ to $f = 10$ (ten times the repeats) the multiplier only grows $0.816 \rightarrow 2.07$, approaching the ceiling $k_1 + 1 = 2.5$. Keyword stuffing has sharply diminishing returns.

Length normalisation (b). The $|d|/\text{avgdl}$ term penalises long documents: a term in a 12-token doc (twice avg) gets a bigger denominator than the same term in a 4-token doc, so short focused documents are not drowned out by long rambling ones. At $b = 0$ length is ignored; at $b = 1$ it is fully normalised; $b = 0.75$ is the usual middle.

Watch out

BM25 is purely lexical: it matches *words*, not meaning. A query for “database error” scores zero against a document about “Postgres connection failure” that never uses those exact tokens. That blind spot is precisely what semantic (cosine) search covers — and why production systems fuse the two (next document).

6 The argument, at a glance

#	Idea	Formula	Value
1	idf (rarity)	$\ln(1 + \frac{N-\text{df}+0.5}{\text{df}+0.5})$	1.204 / 0.357
2	length factor	$1 - b + b \frac{ d }{\text{avgdl}}$	1.375
2	saturated tf	$\frac{f(k_1+1)}{f+k_1(\dots)}$	1.231 / 0.816
3	BM25 of doc	$\sum_t \text{idf} \cdot \text{tf}$	1.773
4	saturation ceiling	$k_1 + 1$	2.5

7 Equation cheat-sheet

Document frequency	$\text{df}(t) = \#\{d : t \in d\}$
idf (BM25)	$\ln(1 + \frac{N-\text{df}+0.5}{\text{df}+0.5})$
Length factor	$1 - b + b d /\text{avgdl}$
BM25 term	$\text{idf} \cdot \frac{f(k_1+1)}{f+k_1(1-b+b d /\text{avgdl})}$
BM25 document	$\sum_{t \in q} \text{BM25 term}(t, d)$
Defaults	$k_1 = 1.5, b = 0.75$

A Reference implementation

```
import math

def idf(N, df):
    return math.log(1 + (N - df + 0.5) / (df + 0.5))

def bm25_term(idf_t, f, dl, avgdl, k1=1.5, b=0.75):
```

```
denom = f + k1 * (1 - b + b * dl / avgdl)
return idf_t * (f * (k1 + 1)) / denom

N, avgdl, dl = 4, 4.0, 6
s_db = bm25_term(idf(N, 1), 2, dl, avgdl)      # 1.482
s_err = bm25_term(idf(N, 3), 1, dl, avgdl)     # 0.291
print(round(s_db + s_err, 3))                  # 1.773
```

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